

Radar HRRP Target Recognition using influence region of samples

Feng Chen, Lan Du, Li Yuan, Zheng Bao

National Lab. of Radar Signal Processing, Xidian University, 710071, China
Chenchen6120@sina.com

Abstract—The k-nearest neighbour(KNN) rule using Euclidean distance is actually the same as template matching method under the maximum correlation coefficient criterion (MCC-TMM), which has been widely used in High resolution rang profiles (HRRPs) based radar automatic target recognition (RATR). The nearest neighbor rule treats each training sample equally without consideration of different recognition performances due to its congregation around the other samples coming from the same class and segregation from those of the rest classes. In this paper, we propose an adaptive method that takes into account the effective influence size of each training sample and the statistical confidence with which the label of each training sample can be trusted. The experimental results confirm the effectiveness of the proposed method.

Index Terms—the k-nearest neighbour(KNN), the nearest neighbour rule, template matching method under the maximum correlation coefficient criterion (MCC-TMM), template matching method(TMM), High resolution rang profiles (HRRPs), radar automatic target recognition (RATR)

I. INTRODUCTION

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community. MCC-TMM, one of the most commonly used RATR algorithms, is identified the nearest neighbor rule using Euclidean distance.

The k-nearest rule is one of the simplest and the most attractive pattern classification algorithms. With only a finite number of training examples, the KNN rule can be viewed as an empirical Bayes classifier based on the estimated a posteriori probability from the k nearest neighbors. Its performance, therefore, relies heavily on the assumption that the a posteriori probability is approximately constant within the neighborhood determined by the k nearest neighbors. With the commonly used Euclidean distance measure, this method is subject to severe bias in a high dimensional input space due to the curse of dimensionality. Due to their high resolution as fine as a decimeter, the dimension of HRRPs is not less than one or two hundreds as the length of a

target is more than 10 meters. Therefore, the direct application of the KNN rule to HRRPs based RATR surely causes the curse of dimensionality. So far, many works have been done to overcome this difficulty, including the flexible metric method by Friedman^[1], the discriminant adaptive method proposed by Hastie and Hibshirani^[2], and the adaptive metric method proposed by Domeniconi et. al.^[3], among others. Although differing in their approaches, the common idea underlying these methods is that they estimated a weighing matrix locally at a query point for computing the distance between it and one training sample. As a result, the computational complexity is high in these methods. The adaptive method proposed by Ji-gangWang^[4] is the most similar one to our approach, but he considered only binary-class classification problem. In our paper, we increase the number of classes to 5 at the same time extract the average HRRP in terms of HRRPs' characteristics to reduce the computational complexity. All of our experiments get much better results than MCC-TMM. In fact, the proposed method can be used to many distance measures using in any decision rule, such as Euclidean distance, L_1 distance, Mahalanobis distance, etc. We just take Euclidean distance as an example for simplicity of comparison in this paper.

Compared to the standard KNN rule, our method has two features: one is that each training sample has an influence region to label its relative distance from other training samples. Roughly speaking, each training sample is associated with an intrinsic influence region, which is defined as the largest sphere centered on the training sample itself that does not include a training sample of the different class. The other is that each influence region is corresponding to a statistical confidence level which is higher than a threshold.

The rest of this paper is organized as follows: in section II, we introduce TMM and the KNN rule and their relation. We then present a new adaptive nearest neighbor rule using the influence region and statistical confidence in section III, wherein we firstly define the probability of error in decisions made by the majority rule based on a finite number of observations, then use it as a criterion for determining the influence region and the statistical confidence that can be associated with each training sample. Finally, we present the results of our experiments and discussions.

This work was partially supported by the National Science Foundation of China (NO.60302009) and the National Defense Preresearch Foundation of China (NO. 413070501).

II. TMM AND THE KNN RULE AND THEIR RELATION

TMM is one of the fundamental methods in pattern classification. It has found wild application in HRRPs based RATR according to various criteria, such as MCC-TMM, adaptive Gaussian classifier (AGC), the maximum likelihood method, etc. Simply speaking, we extract features of HRRPs, and then use them to form the template database. For an arbitrary query point, we search the whole template database and sign it to the class containing the most matching template.

In the following, we take the nearest neighbour rule as an example to explain the KNN rule. Let $D^d = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ denote a set of N labeled training samples and $\mathbf{x}' \in D^d$ is the nearest training sample to a query point $\mathbf{x}$. Then the nearest neighbor rule for classifying $\mathbf{x}$ is to assign it the label associated with $\mathbf{x}'$. It is easy to extend to KNN rule. As one could expect from the name, this rule classifies $\mathbf{x}$ by assigning it the label most frequently represented among the k nearest samples.

When the criterion is the nearest neighbour rule and the distance is defined as Euclidean distance, the TMM is MCC-TMM. The reason is

$$\Delta \mathbf{x} = \|\bar{\mathbf{x}} - \bar{\mathbf{x}}_i\|^2 = \|\bar{\mathbf{x}}\|^2 + \|\bar{\mathbf{x}}_i\|^2 - 2\bar{\mathbf{x}}^T \bar{\mathbf{x}}_i = 2(1 - \bar{\rho}_{\mathbf{x}\mathbf{x}_i}) \quad (1)$$

where $\bar{\mathbf{x}}_i$ and $\bar{\mathbf{x}}$ are normalized training sample and testing point, $\bar{\rho}_{\mathbf{x}\mathbf{x}_i}$ is the normalized correlation coefficient.

III. THE ADAPTIVE METHOD

As what has been discussed above, the k -nearest neighbor is an empirical Bayes classifier; it estimates a posteriori probability $P(\omega_i | \mathbf{x})$ through the k nearest training samples. Theoretically, assuming $P(\omega_i | \mathbf{x})$ keeps constant in a small region around a training sample, the k nearest neighbors can be viewed as i.i.d. from $P(\omega_i | \mathbf{x})$. Therefore, the empirical frequency with which each class ω_i appears among the k nearest neighbor provides an estimate of the a posteriori probability $P(\omega_i | \mathbf{x})$. There are two sources of error in this procedure. One results from the constant a posteriori probability assumption, which states that the class labels of the k nearest neighbors can be approximated as i.i.d.. The other source of error is due to the fact that, even if the constant a posteriori probability assumption holds true, there is still a probability that the empirical majority class differs from the true majority class based on the underlying distribution. In this section, we will address these issues.

A. Probability of Error and Confidence Measure

For simplicity we consider a binary-class $\{+1, -1\}$ classification problem. Let $R \in D^d$ be an arbitrary region in the feature space, wherein there are n samples $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. $p(+1 | \mathbf{x})$ denotes a posteriori probability that a sample $\mathbf{x} \in R$ is classified as $+1$. Assume $p(\pm 1 | \mathbf{x})$ is constant, then the n corresponding labels $Y_1, Y_2, \dots, Y_n$ are i.i.d. We assume that $Y_1, Y_2, \dots, Y_n$ varies according to

Beta-binomial Model that uses the Beta prior model for the binomial distribution. Let us describe Beta-binomial distribution first. Suppose that x varies according to a beta distribution,

$$g(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \alpha, \beta > 0, 0 < x < 1 \quad (2)$$

where $B(\alpha, \beta)$ is the complete Beta function,

$$B(\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, 0 < x < 1 \quad (3)$$

when x is the probability with which a Bernoulli trial occurs, given α and β , the probability distribution of r in n trials is just the probability of Beta-binomial distribution.

$$\begin{aligned} p(r | n, \alpha, \beta) &= \int_0^1 \binom{n}{r} x^r (1-x)^{n-r} g(x) dx \\ &= \int_0^1 \binom{n}{r} \frac{x^{r+\alpha-1}(1-x)^{n-r+\beta-1}}{B(\alpha, \beta)} dx \end{aligned} \quad (4)$$

Without loss of generality, let us assume that R contains n^- samples of class -1 and n^+ samples of class $+1$, with $\delta = n^- - n^+ > 0$. By the majority rule, R will be associated with the class -1 . In practice, $p(+1 | \mathbf{x})$ is unknown, hence whether $p(+1 | \mathbf{x}) > 0.5$ or $p(+1 | \mathbf{x}) < 0.5$ is unknown. If $p(+1 | \mathbf{x}) < 0.5$, it is reasonable for us to assign -1 by the majority rule. But, if $p(+1 | \mathbf{x}) > 0.5$, the class with the maximum a posteriori probability is $+1$ and the classification of R to -1 is a false negative error. According to (4), we get

$$\begin{aligned} p_{err}(n, \delta) &= \frac{\int_{0.5}^1 \binom{n}{r} \frac{x^{r+\alpha-1}(1-x)^{n-r+\beta-1}}{B(\alpha, \beta)} dp}{\int_0^1 \binom{n}{r} \frac{x^{r+\alpha-1}(1-x)^{n-r+\beta-1}}{B(\alpha, \beta)} dp} \\ &= \frac{\int_{0.5}^1 p^{\frac{n-\delta}{2}} (1-p)^{\frac{n+\delta}{2}} dp}{\int_0^1 p^{\frac{n-\delta}{2}} (1-p)^{\frac{n+\delta}{2}} dp} \end{aligned} \quad (5)$$

Since $P_{err}(n, \delta)$ is the probability that the observation is at odds with the true state of nature, $1 - P_{err}(n, \delta)$ is the probability that the observation agrees with the true state of nature. We therefore define

$$\alpha(n, \delta) = 1 - p_{err}(n, \delta) = \frac{\int_0^{0.5} p^{\frac{n-\delta}{2}} (1-p)^{\frac{n+\delta}{2}} dp}{\int_0^1 p^{\frac{n-\delta}{2}} (1-p)^{\frac{n+\delta}{2}} dp} \quad (6)$$

to be the confidence level of our decision.

Fig.1 shows the variation of confidence level with n and δ . To keep the confidence level above a threshold, say 0.8, there must be more than 54 templates of class -1 to classify R as -1 for the total number of templates n is 100, more than 28 for $n = 50$, more than 12 for $n = 20$. Obviously, the relation between $\alpha(\delta, n)$ and n, δ is nonlinear. For a constant n , $\alpha(\delta, n)$ increases with

augment of δ ; for a constant δ , it decreases with augment of n . We can safely get the idea confidence level increases when there is more samples from the same class and less ones from the other class.

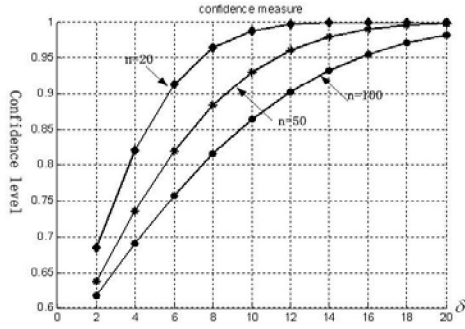


Figure 1. Variation of confidence level with n and δ

B. The Adaptive Rule using Confidence Level and Influence Region

Unlike the traditional MCC-TMM, our adaptive method based on Euclidean distance starts with constructing an influence region for each sample or calculating an influence region for each sample above a certain confidence level. Generally, we extract single rang profile or average rang profile within a small azimuth angle rang for MCC-TMM. Its deficiency is not only lack of using the similarity characteristic between rang profiles abutting on each other in time domain, but also the idea treating every sample equally in despite of it already exists on the class boundary. Figure 2 illustrates the situation for a simple example, where the data in each class are uniformly distributed in a half-square, and perfectly separately by a vertical line. The sphere denoting the nearest neighbour rule extends to a boundary sample that will classify the black dot belonging to class -1 to class +1.

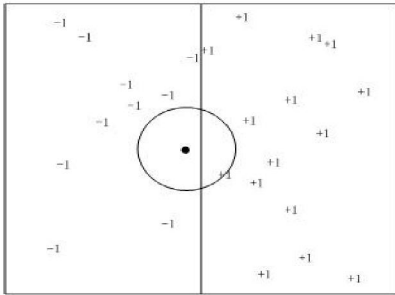


Figure 2. A failed example of the nearest rule on the decision boundary

The influence region proposed here is a sphere as large as possible centered on a sample without enclosing any sample from the different class. We denote the radius of the sphere as r_i and count the number of training samples that fall inside the sphere and compute the statistical confidence level according to (6). Based on the value of the computed confidence level, there are two possibilities. If the confidence level is above a preset threshold, e.g., 75%, then we keep the influence region as it is. Otherwise, we would enlarge the sphere to include more training samples until its confidence level reaches the threshold. In

the second case, the radius is set to the radius of the sphere that achieves the desired confidence level, and so is the confidence level.

After finishing constructing an influence sphere for each training sample, we define a new distance,

$$d_{new}(\mathbf{x}, \mathbf{x}_i) = \frac{d(\mathbf{x}, \mathbf{x}_i)}{r_i} \quad (7)$$

where $d(\mathbf{x}, \mathbf{x}_i)$ is the Euclidean distance between testing point $\mathbf{x}$ and training sample $\mathbf{x}_i$. Then we find the k -nearest neighbor according to this new distance $\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots, \mathbf{x}_{(K)}$, and classify $\mathbf{x}$ to

$$f(\mathbf{x}) = \text{sgn}(\alpha_{(i)} Y_{(i)}) \quad (8)$$

According to this new definition of distance, every testing point inside the influence sphere of a training sample has a distance to the center smaller than 1, while outside bigger than 1. In addition, as one can easily imaging, a neighbour that happens to be close to decision boundary would have less weight because of the lower confidence level.

IV. EXPERIMENTS AND DISCUSSION

We simulate radar backscattering data of five aerial targets^[5], as Table I shows. Because the first three targets are much bigger than the rest, we shrink them to half in order to increase the recognizing difficulty. As these five aerial targets are all symmetrical in geometry, we only simulate azimuth $0 \sim 180^\circ$.

TABLE I PLANE TARGETS AND RADAR REFERENCE

radar reference	center frequency		5520 MHz
	PRF		400 Hz
	sampling frequency		400 MHz
	signal bandwidth		400 MHz
plan type	length (m)	width (m)	proportion
Tu-16	34.80	33.00	2: 1
B-52	49.50	56.40	2: 1
B-1B	44.80	23.80	2: 1
Tornado	16.72	13.91	1: 1
Mig-21	15.76	7.15	1: 1

In our first experiment, single rang profile at intervals of 0.2° azimuth angle is chosen as feature to establish template database. Table 2 shows the comparison in detail between MCC-TMM and our adaptive method. There are 10 kinds of combinations to find two kinds of targets from the five targets. Table II shows for each kind of combination our adaptive method has a higher recognition rate than MCC-TMM with an average gain about 3 percents. As the number of targets increases, the recognition rate decrease, but our adaptive method gets much more increase comparing to MCC-TMM. It seems our method suits to solve multi-class classification problems. Table III shows another experimental results using average rang profile within 5° azimuth angle rang as feature vector. The number of templates is 25 times less

TABLE II COMPARISON OF RECOGNITION RATE USING SINGLE PROFILE AS FEATURE BETWEEN MCC-TMM AND THE PROPOSED METHOD. FOR CONVENIENCE, T1 DENOTES THE TU-16 PLAN, T2 DENOTES B-52, T3 DENOTES B-1B, T4 DENOTES TORNADO, T5 DENOTES MIG-21.

		Recognition rate (%)										Average recogniti-on rate (%)
2 classes	classes	T1vsT2	T1vsT3	T1vsT4	T1vsT5	T2vsT3	T2vsT4	T2vsT5	T3vsT4	T3vsT5	T4vsT5	
	MCC-TMM	93.06	92.22	94.44	96.94	93.61	92.50	93.33	92.78	87.22	88.89	92.50
	ADAPT METHOD	96.39	94.72	98.06	98.61	96.39	97.22	97.78	95.28	88.33	91.94	95.47
3 classes	classes	T1vsT2 vsT3	T1vsT2 vsT4	T1vsT2 vsT5	T1vsT3 vsT4	T1vsT3 vsT5	T1vsT4 vsT5	T2vsT3 vsT4	T2vsT3 vsT5	T2vsT4 vsT5	T3vsT4 vsT5	
	MCC-TMM	88.33	90.00	90.56	88.70	86.67	88.89	88.15	85.19	86.48	83.15	87.61
	ADAPT METHOD	92.56	95.37	96.11	92.78	88.70	92.78	93.70	89.63	92.78	86.85	92.13
4 classes	classes	T1vsT2vsT3vsT4		T1vsT2vsT3vsT5			T2vsT3vsT4vsT5		T3vsT4vsT5vsT1			
	MCC-TMM	86.11		84.17			81.53		82.78			83.65
	ADAPT METHOD	91.25		88.33			87.92		92.36			89.97
5 classes	classes	T1vsT2vsT3vsT4vsT5										
	MCC-TMM	81.33										81.33
	ADAPT METHOD	86.89										86.89

than that of single rang profile one. Therefore, the computational complexity is low. The recognition rate using adaptive method also increases more than that using MCC-TMM as the target number increases.

TABLE III COMPARISON OF AVERAGE RECOGNITION RATE USING AVERAGE PROFILE AS FEATURE BETWEEN MCC-TMM AND THE PROPOSED METHOD

	2 classes	3 classes	4 classes	5 classes
MCC-TMM	86.21	76.69	64.65	64.00
ADAPT METHOD	87.41	77.50	66.84	68.11
increase	1.21	0.81	2.19	4.11

Here we have discussions on three topics.

1) Various approaches have been developed in order to handle multi-class classification problems with binary-class classifier, including one-against-rest classifiers, one-against-one classifiers and hierarchies or trees of binary-class classifiers. The influence region of a sample is used to label the relative distance from it to the samples coming from the same class or the other classes; obviously, it isn't suitable to regard different classes as a new bigger class. Moreover, the complexity of training influence region increases if one-against-rest classifiers are used. We choose one-against-one classifiers in our experiments, and all the results given are based on them.

2) The adaptive method searches all the N samples to decide the influence region for each sample, therefore, we need do N^2 loops. Particularly, when the feature extracted is single rang profile, it requires high computational complexity and large storage. However, the operation of training influence regions is limited in the template database that does not need any testing point. Therefore, we can calculate the influence regions for each classifier

and store them in the computer beforehand, and then use them directly for testing. In addition, we actually do HRRPs based RATR after detection of a target, which has had already provide us an approximate azimuth angle of the target with an attainable precision of $\pm 15^\circ$. Accordingly, for a query point, we may test it around the corresponding azimuth angle rang. Experimental results show that, in this way, we improve the recognition rate while reducing computational complexity. Table III shows the results obtained with no azimuth angle estimation, and we notice these are much smaller than those in table II with azimuth information.

3) In experiments, all the KNN is the nearest neighbor rule. One usually uses methods such as cross-validation to pick the best value for the value of k in KNN, and it is a hard work requiring long in time and high in computational complexity. We find the nearest rule often achieves the best or approximately the best recognition performances in HRRPs based RATR.

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